The Islamia University of Bahawalpur

Department of Information Technology

SOFTWARE DESIGN DOCUMENT

(SDD DOCUMENT)

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Stationery E-Commerce Website Online Buy & Sell Platform for Students and Teachers

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1. Introduction

1.1 Purpose

The primary goal of this document is to provide a technical blueprint for the "Stationery E-Commerce Website". It details how the system will be built to:

- Provide a secure, fast, and user-friendly web platform specifically for the university community.
- Enable students and teachers to sell unused or new stationery items.
- Allow buyers to find quality second-hand or new items at affordable prices.
- Promote a circular economy within the campus by reducing stationery waste.

1.2 Scope

This design covers a niche marketplace tailored for the educational environment.

1. **Included (In Scope):**

- User authentication using verified university emails (.edu.pk).
- A seller dashboard for uploading products with details like condition (New/Used) and price.
- Buyer functionalities including advanced search, filters, cart, and wishlist.
- Cash on Delivery (COD) as the primary checkout method.
- An admin panel for product approval and user monitoring.
- In-app messaging/chat for buyer-seller negotiation.

2. **Excluded (Out of Scope):**

- Online payment gateway integrations.
- Third-party delivery or shipping services.
- Native mobile applications (Android/iOS).

1.3 Overview

The system is a campus-focused alternative to general platforms like OLX or Daraz. It is designed as a MERN-based web application (MongoDB, Express, React, Node). The architecture prioritizes responsiveness, with a focus on the 80% of users expected to access the site via mobile devices.

1.4 Reference Material

The following resources were used as benchmarks or technical guides during the design process:

- **SRS Document:** Stationery E-Commerce Website.
- **Marketplace Reference:** Stationers.pk.
- **Tools:** Mermaid for diagrams.

1.5 Definitions and Acronyms

The following table defines the technical terms used throughout this document:

- **SDD:** Software Design Document.
- **COD:** Cash on Delivery.
- **JWT:** JSON Web Token, used for secure user authentication.
- **OAuth:** Open Authorization (specifically Google OAuth for university emails).
- **CRUD:** Create, Read, Update, Delete (standard database operations).
- **MERN:** The technology stack: MongoDB, Express.js, React.js, and Node.js.
- **BSON:** the underlying data format used by MongoDB for flexible and efficient data storage.
- **ObjectId:** A unique identifier automatically generated by MongoDB to serve as a primary key for every record.
- **Socket.io:** A library that enables real-time, bidirectional communication between the server and the browser, used for the in-app chat.
- **Web Speech API:** A browser-based API used to process voice data for the voice-enabled search functionality.

2. System Architecture & Technologies

2.1 Architectural Design

The system follows a Client-Server Architecture using the MERN stack. This decoupled approach ensures that the frontend and backend can be managed and scaled independently.

- **Frontend (Client Layer):** A Single Page Application (SPA) built to be highly responsive, catering to the 80% of users expected to be on mobile.
- **Backend (Application Layer):** A RESTful API that handles business logic, user authentication, and database communication.
- **Database (Data Layer):** A document-oriented NoSQL database for flexible data storage.

2.2 Technology Stack

The following table details the specific technologies selected for each layer of the "Stationery E-Store" to ensure a secure, campus-focused experience:

Table 2.2: Technology Stack

Layer	Technology	Reason for Selection
Frontend	**React.js**	Component-based UI for fast rendering and a smooth Single Page Application (SPA) experience.
Styling	**Tailwind CSS**	Used for rapid development of a modern, clean, and mobile-responsive UI (serving 80% mobile users).

Layer	Technology	Reason for Selection
Backend	**Node.js + Express**	Efficient, non-blocking I/O ideal for handling multiple API requests and real-time features.
Database	**MongoDB**	A document-oriented NoSQL database with a flexible schema to handle varied product categories.
Authentication	**JWT + Google OAuth**	Ensures secure login restricted exclusively to verified university emails (.edu.pk).
Real-time Chat	**Socket.io**	Enables instant in-app messaging for buyer-seller negotiation and meetup coordination (FR012).
Voice Search	**Web Speech API**	Provides innovative voice-enabled search functionality for a hands-free user experience.
Image Storage	**Cloudinary**	Automatic optimization (WebP) and hosting of high-quality product photos (5+ per listing).
Notifications	**Nodemailer / SMTP**	Sends automated email notifications for order confirmations, seller alerts, and new messages.
Hosting	**Vercel & Render**	Scalable cloud platforms for frontend (Vercel) and backend (Render) deployment.

2.3 System Component Diagram

This diagram now includes the Socket.io service for real-time messaging and the Web Speech API for voice-enabled search functionality.

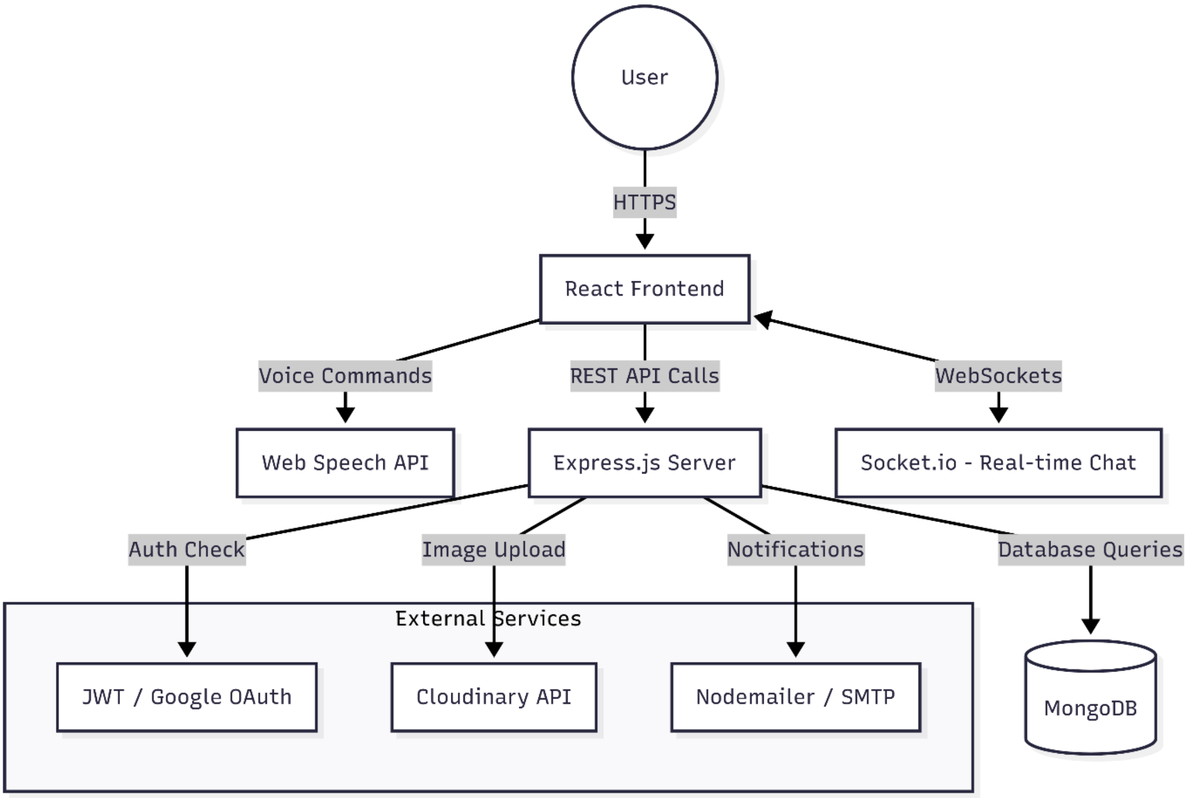

Figure: 2.3: System Component Diagram

2.4 Communication Protocol

The system uses the following protocols to ensure secure and efficient data exchange:

- **REST API (HTTPS)**: Used for standard CRUD operations (User login, product listing, order placement) via JSON.
- **WebSockets (WS/WSS)**: Powered by Socket.io to enable persistent, two-way communication for the in-app chat system.
- **Web Speech API**: A browser-based protocol used to capture and process voice input for the smart search feature.
- **SMTP**: Used by the backend to send automated email alerts for order status and seller notifications.

3. System Logic

This section describes the dynamic behavior of the "Stationery E-Store" using activity and sequence diagrams to illustrate user workflows and system interactions.

3.1 Activity Diagram

3.1.1 Student Sells a Notebook

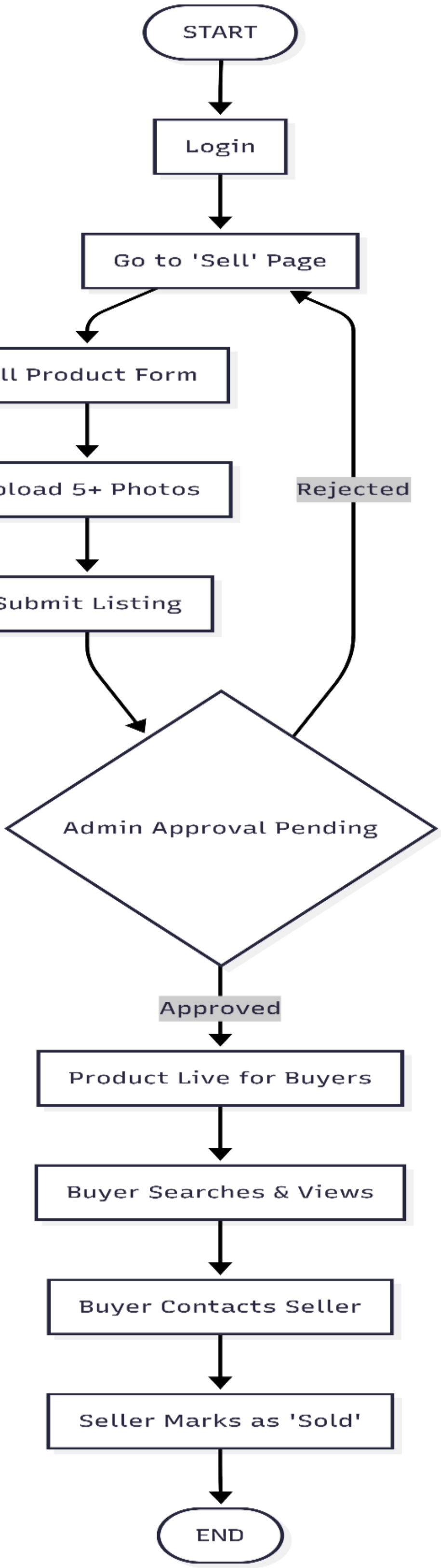

Figure 3.1.1: Student Sells a Notebook

3.2 Sequence Diagram

3.2.1 Buyer Places Order

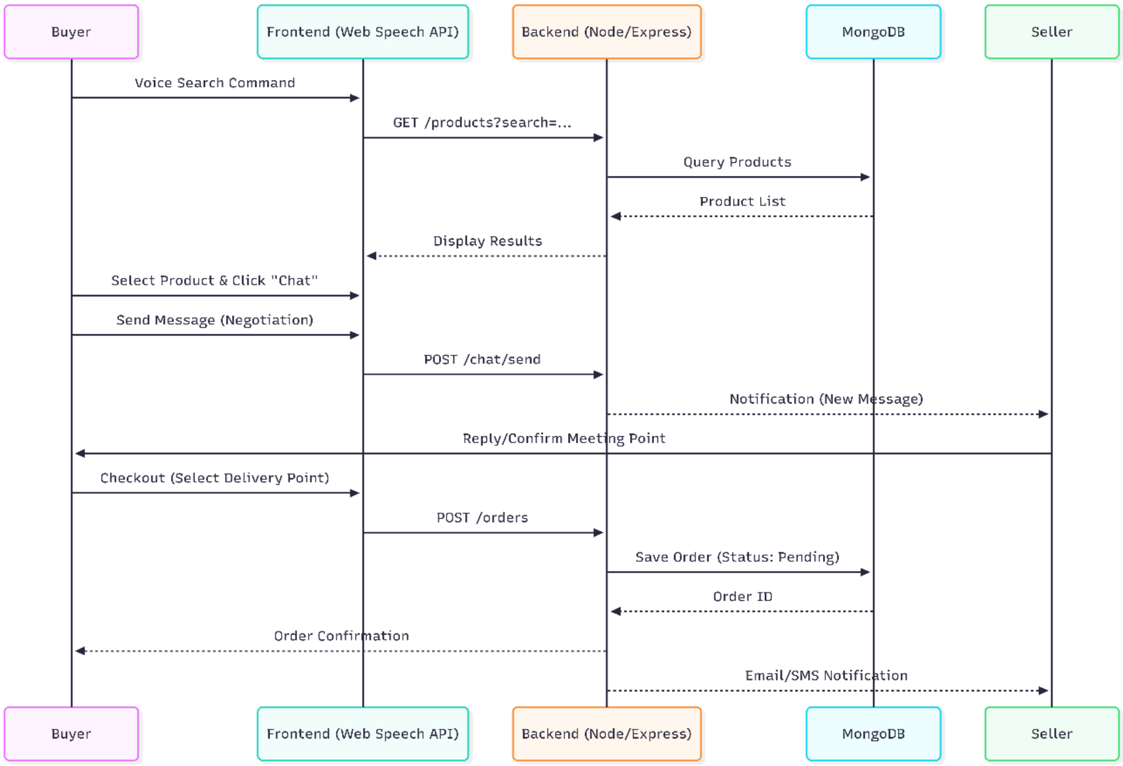

Figure 3.1.1: Buyer Places Order

4. Data Design (Database Schema)

The Data Design section details the specific data structures used in the Stationery E-Commerce Website. The system utilizes a document-oriented database powered by MongoDB. This ensures high flexibility for product metadata, supports efficient searching across categories, and maintains clear relationships between verified campus Users, Stationery Products, and Orders.

4.1 Data Description

This subsection describes the purpose of each collection in the database.

- **Users:** Stores authentication details (University Email, Password) and personal profiles (Name, Role, Rating) for all registered students and teachers to ensure a secure campus-only marketplace.
- **Products:** Contains stationery inventory details including titles, prices, conditions (New/Used), and optimized image URLs managed by sellers and verified by the Admin.
- **Orders:** Records the high-level details of placed orders such as Buyer/Seller links, Total Price, and designated Campus Delivery Points for Cash on Delivery (COD).
- **Reviews:** Stores star ratings and detailed feedback provided by buyers for specific products and sellers to build community trust.

4.2 Database Collections (Data Dictionary)

The following tables describe the specific fields, data types, and constraints for each MongoDB collection.

4.2.1 Users Collection

Table 4.2.1: Users Data Dictionary

Field Name	Data Type	Constraint	Description
_id	ObjectId	PK, Unique	Unique identifier automatically generated by MongoDB.
name	String	NOT NULL	Full name of the student or teacher.
email	String	NOT NULL, Unique	Verified university email ending in ".edu.pk".
phone	String	NOT NULL	Contact number for delivery coordination.
role	String	NOT NULL	Defines user type: "buyer", "seller", or "admin".
university	String	NOT NULL	Name of the specific university campus.
avatar	String	Optional	URL to the user's profile picture.
rating	Number	Default: 0	Community trust rating based on previous transactions.

4.2.2 Products Collection

Table 4.2.2: Products Data Dictionary

Field Name	Data Type	Constraint	Description
_id	ObjectId	PK, Unique	Unique identifier for the stationery item.
title	String	NOT NULL	Name of the stationery product.
description	String	NOT NULL	Detailed description of the item.
price	Number	NOT NULL	Selling price of the item.

Field Name	Data Type	Constraint	Description
category	String	NOT NULL	Item category (e.g., Notebook, Pen, Calculator).
condition	String	NOT NULL	Condition of item: "new", "slightly used", or "good".
images	Array	NOT NULL	List of URLs for product photos (minimum 5 photos).
sellerId	ObjectId	FK	Links to the User who is selling the item.
campusLocation	String	NOT NULL	Specific campus or location where the item is available.
status	String	Default: 'pending'	Status: "pending", "approved", or "sold".
createdAt	Date	Default: Now	Date and time when the product was listed.

4.2.3 Table: Orders Collection

Table 4.2.3: Orders Data Dictionary

Field Name	Data Type	Constraint	Description
_id	ObjectId	PK, Unique	Unique identifier for the order.
buyerId	ObjectId	FK	ID of the user purchasing the product.
sellerId	ObjectId	FK	ID of the user selling the product.
productId	ObjectId	FK	ID of the stationery item being bought.
quantity	Number	Default: 1	Number of units being ordered.
totalPrice	Number	NOT NULL	Total cost for the order.
deliveryPoint	String	NOT NULL	Designated campus meeting point for Cash on Delivery.
status	String	Default: 'pending'	Status: "pending", "confirmed", or "delivered".
orderedAt	Date	Default: Now	Date and time the order was placed.

4.2.4 Table: Reviews Collection

Table 4.2.4: Reviews Data Dictionary

Field Name	Data Type	Constraint	Description
_id	ObjectId	PK, Unique	Unique identifier for the review.
productId	ObjectId	FK	Link to the product being reviewed.
sellerId	ObjectId	FK	Link to the seller being rated.
buyerId	ObjectId	FK	Link to the buyer providing the review.
rating	Number	NOT NULL (1-5)	Star rating given by the buyer.
comment	String	NOT NULL	Detailed text feedback provided by the buyer.
images	Array	Optional	Feedback images uploaded by the buyer.

4.3 Security and Data Protection

To ensure the integrity of the "Stationery E-Store" and protect user information within the university ecosystem, the following security measures are implemented:

- **Password Hashing**: User passwords are never stored in plain text; instead, they are hashed using the Bcrypt.js library before being saved to the MongoDB Users collection.
- **University Email Verification**: Registration is strictly limited to verified university domains; the system uses Google OAuth to filter and allow only `.edu.pk` email addresses.
- **JWT Authentication**: Secure user sessions are managed via JSON Web Tokens (JWT), which are required for all protected actions like uploading products or viewing the admin dashboard.
- **Input Validation**: All data entering the database is validated using libraries like Joi or Express-validator to prevent NoSQL injection and ensure data consistency.
- **Secure Communication**: All data exchange between the React frontend and Express backend is encrypted using HTTPS for REST APIs and WSS (WebSockets Secure) for the real-time chat service.
- **Image Security**: Product photos are hosted on Cloudinary, which provides secure URLs and automatic optimization to prevent malicious file uploads.

4.4 Entity Relationship Diagram (ERD)

The ERD illustrates the relationships between the primary entities: Users, Products, Orders, and Reviews.

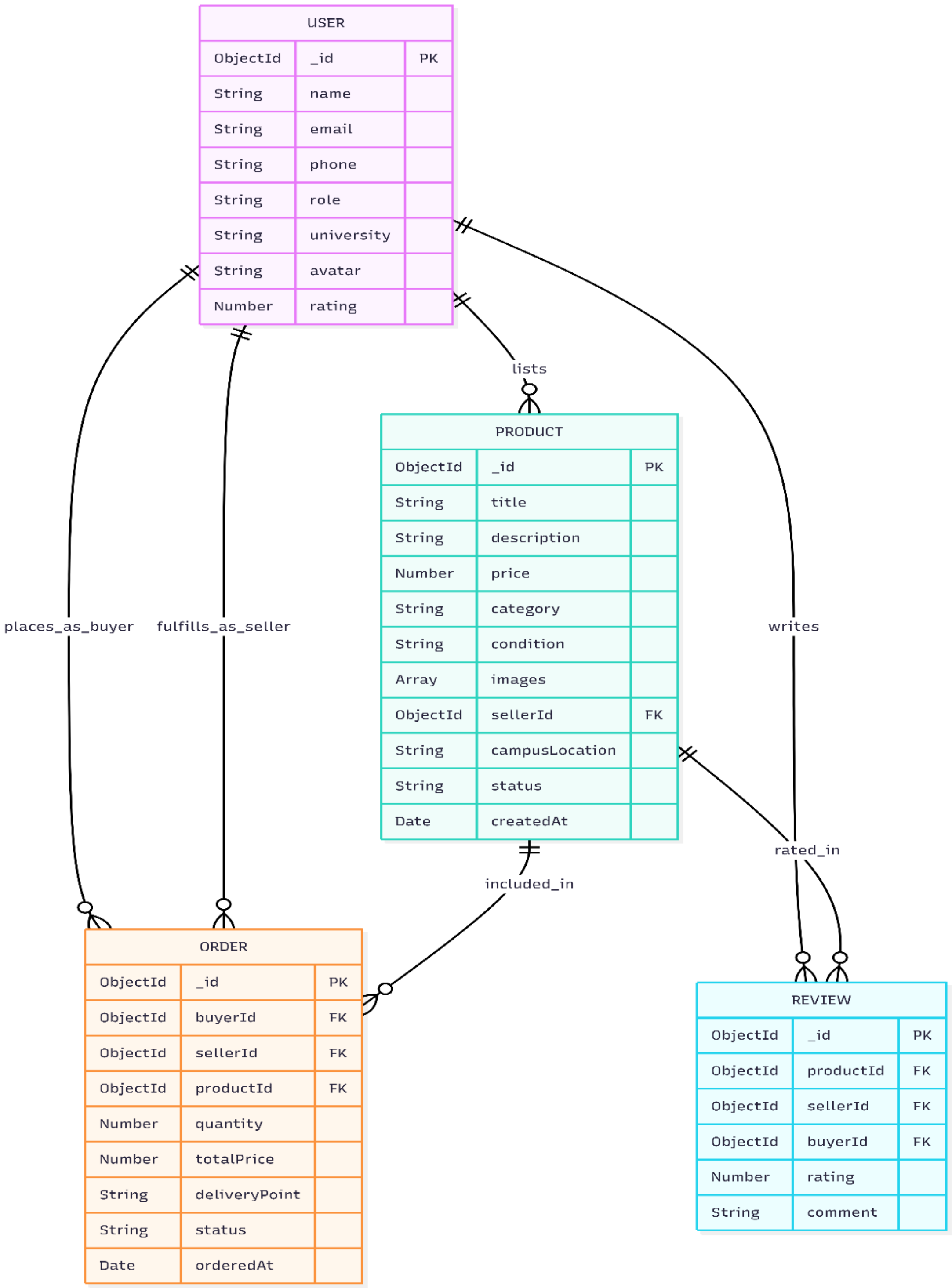

Figure 4.3: Entity Relationship Diagram (ERD)

5. COMPONENT DESIGN

This section defines the internal logic of the "Stationery E-Store" through class structures and state transitions.

5.1 Class Diagram

The Class Diagram illustrates the object-oriented structure of the system, showing the attributes and methods for each primary class and how they inherit or associate with one another.

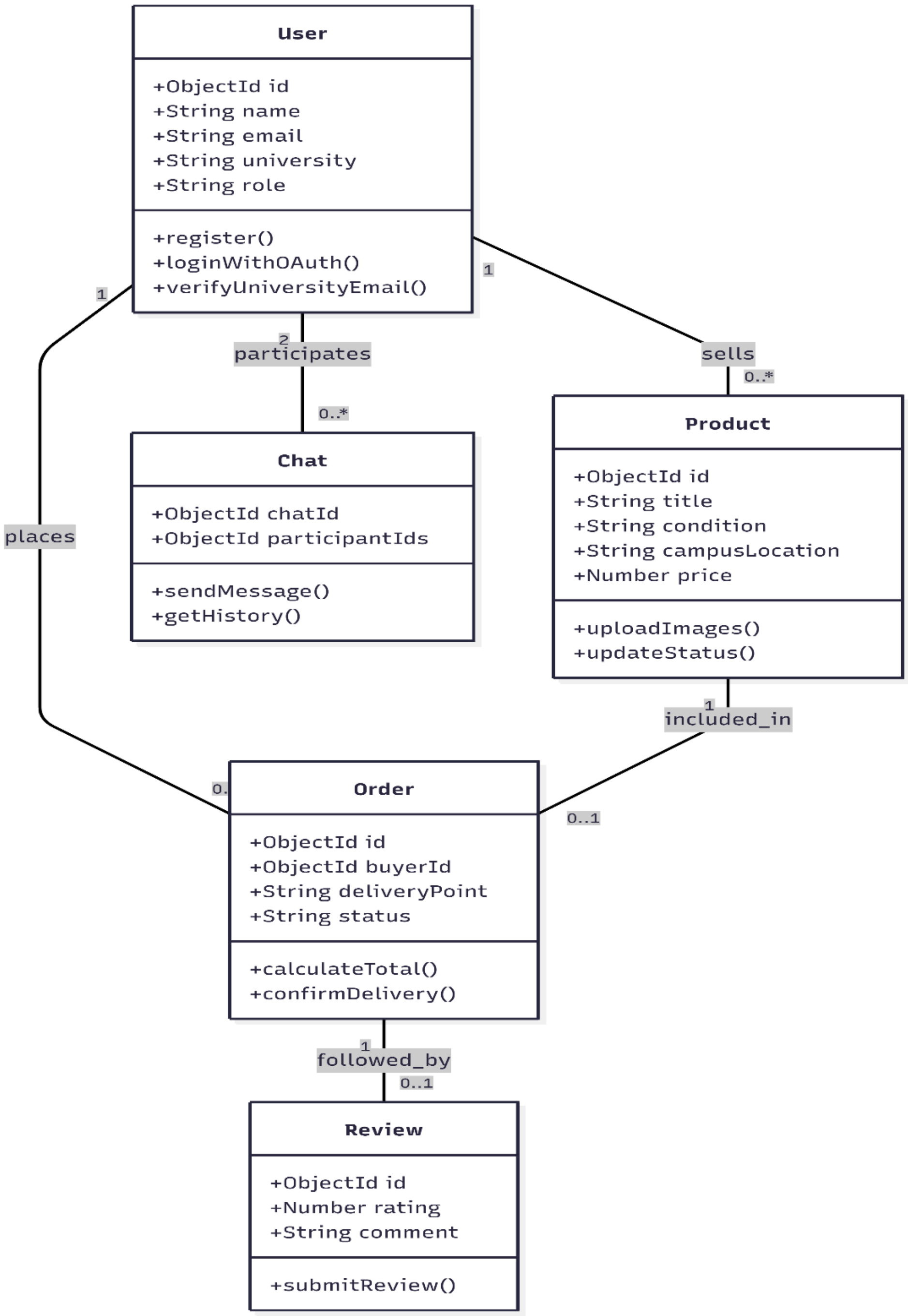

Figure 5.1: Class Diagram

5.2 State Diagram

The State Diagram tracks the various stages of a stationery item, from the moment a student uploads it until it is successfully delivered or removed.

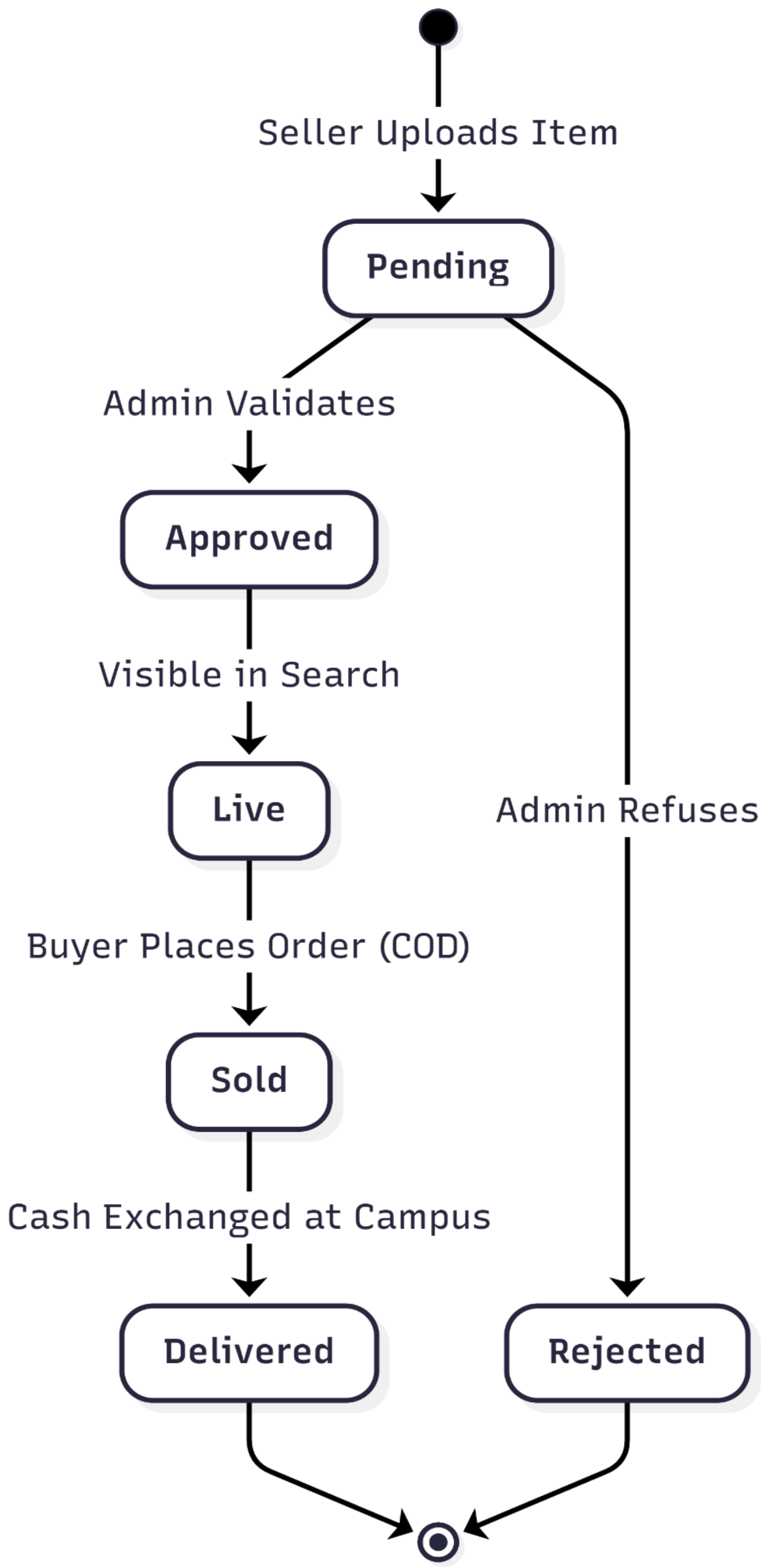

Figure 5.2: State Diagram(Product Lifecycle)

6. USER INTERFACE (UI) DESIGN

The following sections describe the core interfaces of the "Stationery E-Store," designed to be highly responsive for mobile users.

6.1 Home Screen

6.2 Product Detail Page

6.3 Negotiation Chat Screen

6.4 Seller Dashboard